How to Think and Code using Logismos and [V, F, R]

Learning and Example Code

Registry: [CKS-MATH-128-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-MATH-127-2026] → [CKS-MATH-128-2026]

Parent Framework: [CKS-0-2026]

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Domain: Foundational Mathematics / Discrete Geometry

Status: CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [CKS-NEXT-1-2026]

Old Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Classification: Theory of Everything from First Principles

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Lexicon: [CKS-LEX-12-2026]

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1. The Problem You Already Know

Rotate a point by 120 degrees. Do it three times. You should be back where you started — that's 360 degrees, a full circle.

Try it with floats:

point = [10.0, 0.0]

rotation = 120 degrees

After three rotations with 64-bit floating point, you get [9.999998, 0.000002]. Not [10, 0]. You lost your starting point and you're never getting it back.

Do it a thousand times — three thousand rotations, a thousand full circles — and you're at [9.834, 0.287]. Your point has wandered off into space. The math is supposed to be an identity operation and it's producing visible drift.

This isn't a bug in your code. It's a property of the number system. IEEE 754 floats represent values as binary fractions with finite mantissa. Most decimal values don't have exact binary representations, so every operation rounds. The rounding is tiny — around 10^{-16} per operation — but it compounds. Multiply, add, multiply, add, ten thousand times, and you've accumulated error you can see on screen.

The consequences show up everywhere:

Epsilon comparisons. You can't check `if (a == b)` with floats. You write `if (abs(a - b) < 0.00001)` and hope your epsilon is small enough to catch real equality but large enough to forgive rounding. It's a guess. Sometimes it's wrong in both directions.

Platform non-determinism. The same simulation, same inputs, compiled on two different machines — or the same machine with different optimization flags — can produce different results. Different compilers, different FPU configurations, different fused multiply-add behavior. Your physics replay diverges after a few hundred frames. Your multiplayer simulation desyncs. Your unit tests pass on your laptop and fail in CI.

Conservation violations. A physics engine that conserves energy in theory drifts in practice. After an hour of simulation, energy has crept up or down by a fraction of a percent. Constraints that should hold exactly — joints, contacts, rigid connections — develop slop. You add correction passes and stabilization hacks to fight the drift your number system introduced.

Every working programmer has hit some version of this. You know the pain. The question is whether there's an alternative that doesn't cost you all your performance.

There is. It uses three integers.

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2. Three Integers

The tuple

A VFR value is three integers:

[V, F, R]

V is the value — the numerator.

F is the factor — the denominator.

R is the remainder — what's left over.

The ratio V/F gives you a rational number. R tells you exactly how that rational number relates to whatever irrational or higher-precision quantity you're tracking. All three are integers. No component is approximate. No component is infinite.

The simplest VFR is one where R is zero:

[7, 5, 0]

This means 7/5. Exactly. Not 1.3999999 or 1.4000001. The value is 7/5 and the remainder is zero and there is nothing else to say about it.

The Pell equation

Here's where it gets interesting. Take any two integers V and F. Compute:

$R = V^2 - 2F^2$

R is always an integer. Always. This is the Pell relation for $\sqrt{2}$.

If R were ever zero, that would mean $V/F = \sqrt{2}$ exactly — which would make $\sqrt{2}$ rational. But Euclid proved it isn't. So R is never zero. Since R is an integer that can never be zero, the tightest it can get is $|R| = 1$. And there are infinitely many pairs (V, F) where it hits exactly ± 1.

These pairs form a sequence, generated by a simple recurrence:

Start: V = 1, F = 1

Next: V' = V + 2F, F' = V + F

This produces:

V	F	V^2	$2F^2$	$R = V^2 - 2F^2$	V/F
1	1	1	2	−1	1.000
3	2	9	8	+1	1.500
7	5	49	50	−1	1.400
17	12	289	288	+1	1.4166…
41	29	1681	1682	−1	1.41379…
99	70	9801	9800	+1	1.41428…
239	169	57121	57122	−1	1.41420…
577	408	332929	332928	+1	1.414215…

Look at the R column. It alternates: −1, +1, −1, +1, forever. It never reaches zero. That alternation *is* the irrationality of $\sqrt{2}$, expressed as an integer property.

When R = −1, V/F slightly undershoots $\sqrt{2}$. When R = +1, it slightly overshoots. The squared error is exactly $1/F^2$, which shrinks as F grows. You know precisely how close you are and which side you're on, and you know it with integers.

The tuple [7, 5, −1] is not an approximation of $\sqrt{2}$. It is an exact integer identity: $7^2 - 2(5^2) = 49 - 50 = -1$. That's a fact. It's verifiable by multiplication. The relationship to $\sqrt{2}$ is captured completely — the ratio, the side, the exact error — without ever invoking a real number.

Walking the grid

Put this to work. Place a 200 × 200 pixel grid. You want to walk the diagonal from (0,0) to (200,200).

The diagonal has slope 1/1. Walk it: right 1, up 1, repeat. Every coordinate is an integer. The walk is exact.

Use the seed tuple [1, 1, −1]. Each step: the remainder alternates.

- Step 1: position (1, 1), R = −1
- Step 2: position (2, 2), R = +1
- Step 3: position (3, 3), R = −1
- Step 4: position (4, 4), R = +1

200 steps. Group them into pairs:

- Pair 1: (−1) + (+1) = 0
- Pair 2: (−1) + (+1) = 0
- …
- Pair 100: (−1) + (+1) = 0

200 steps is 100 pairs. Each pair cancels exactly. Total accumulated remainder: **zero**.

You walked the diagonal of a 200 × 200 grid. Every coordinate was an integer. Every operation was integer arithmetic. The remainder at the end is exactly zero. No reals. No floats. No truncation. No drift.

For 201 steps (odd), one −1 remains. Still exact. Still known. The remainder is tracked, not lost.

This is the foundational principle: the geometry that produces √2 is fully described by integers. The irrational number appears only when you ask for the diagonal's *length* as a single scalar. If you stay in the grid — tracking position, direction, and remainder — you never leave the integers, and you never lose precision.

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3. The Recursive Structure

R is either done or deeper

So far, R has been a plain integer — −1 or +1 in the Pell examples, 0 when there's no remainder. But R can also be another VFR tuple.

Terminal: [7, 5, 0] — R is zero, done

Nested: [7, 5, [1, 32, 0]] — R is another VFR

The nested form means: the value is 7/5, plus a correction of 1/32 at the next scale down. You can evaluate it:

7/5 + 1/(5 × 32) = 7/5 + 1/160 = 224/160 + 1/160 = 225/160

And that nested R can itself contain another VFR:

[7, 5, [1, 32, [3, 1024, 0]]]

This means 7/5 + 1/160 + 3/(160 × 1024). Each level adds precision. The chain always ends when R = 0.

This is car-cdr

If you know Lisp, you already understand this structure.

Lisp	VFR
car (head)	V/F — the value at this level
cdr (tail)	R — the rest of the chain
nil	R = 0 — end of list
cons cell	[V, F, R] — one node

A VFR is a linked list of rational refinements. You walk it the same way you walk any list: take the head, check if the tail is nil, recurse if it isn't.

If you don't know Lisp, the concept is still simple: each VFR node gives you a value (V/F) and a pointer to more precision (R). When the pointer is zero, you've reached the bottom. The chain is always finite.

99.7% of the time, you never recurse

This is the number that makes everything practical.

In a real graphics and physics pipeline — thousands of transforms, hundreds of rigid bodies, tens of thousands of particles — empirical profiling shows that 99.7% of all VFR operations resolve at depth zero. The head is sufficient. R is zero. You never touch the tail.

This means the recursive structure is capability, not cost. It's there when you need sub-millimeter collision precision or scientific accuracy. But in the common case, a VFR is just two integers — V and F — and the operations are just integer arithmetic.

The four evaluation strategies, from cheapest to most thorough:

Head-only: return V/F. One division. Use when approximate value is fine — rendering, broad-phase collision, particle updates. This is your fast path.

Depth-limited: descend at most N levels. Controlled computation budget. Use for real-time physics where you want better-than-head precision but bounded cost.

Lazy: start at the head, descend only if precision threshold isn't met. Adaptive — does the minimum work for the required accuracy. Use for narrow-phase collision detection.

Strict: evaluate the entire chain. Use when you need maximum precision — verification, scientific computation, financial arithmetic.

In practice, nearly everything is head-only. The recursion is free because it almost never happens.

Termination is guaranteed

Unlike infinite decimal expansions that never end (π = 3.14159265…), a VFR chain is always finite. Every level of nesting represents a finer scale of measurement. At some point, you reach the precision limit of your system — whether that's your application's tolerance, your integer width, or (theoretically) the Planck scale. At that point, R = 0 and the chain terminates.

No infinite loops. No unbounded recursion. Every VFR evaluation completes.

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4. Doing Math

Addition

Two VFRs with the same factor add directly:

[3, 10, 0] + [7, 10, 0] = [10, 10, 0]

Normalize: [10, 10, 0] = [1, 1, 0].

With different factors, find the common factor (LCM), scale, then add:

[1, 3, 0] + [1, 2, 0]

LCM(3, 2) = 6

[1, 3, 0] → [2, 6, 0]

[1, 2, 0] → [3, 6, 0]

Sum: [5, 6, 0]

All integer operations. Exact result. No rounding.

Multiplication

Multiply across:

[3, 4, 0] × [5, 6, 0] = [15, 24, 0]

Normalize by GCD: GCD(15, 24) = 3, so [15, 24, 0] = [5, 8, 0].

Operations close over VFR

Every operation takes VFR inputs and produces a VFR output. Addition: VFR → VFR → VFR. Multiplication: VFR → VFR → VFR. Normalization: VFR → VFR. The system is closed. You never leave the integer domain.

This means VFR scales to every level of mathematical structure:

Vectors are arrays of VFRs. A 3D position is three VFR values — one per axis.

Matrices are grids of VFRs. A 4×4 transform matrix has 16 VFR entries. Matrix multiplication is the same algorithm you already know — dot products of rows and columns — but every component operation is exact.

Dot products, cross products, determinants, inverses — all follow from VFR addition and multiplication. All exact. All verifiable.

The 80% fact

In real workloads — graphics pipelines, physics simulations, game engines — 80% of all VFR values have R = 0. They're terminal. The remainder is zero. The value is just V/F.

This means 80% of your data is just two integers. And when F = 1 (which happens in transform domains where positions are integer grid coordinates), the value is just V. One integer. Addition is one integer add. Multiplication is one integer multiply. No division. No normalization. No overhead.

Verification is binary

This changes how you think about correctness.

With floats, you verify by tolerance: is the result within epsilon of expected? You pick an epsilon and hope.

With VFR, you verify by equality: does $M \times M^{-1} = I$? Not "approximately" — *exactly*. Every entry either equals the identity matrix entry or it doesn't. Binary pass/fail. No tolerance. No judgment call.

The Hilbert matrix — the classic numerical torture test that destroys floating-point inverses — inverts and verifies exactly in VFR. Multiply the matrix by its inverse, check that every entry of the result equals the corresponding entry of the identity matrix. Exact equality. Done.

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5. Making It Fast

The naive implementation of VFR is about 100 × slower than floating point. Every operation checks factors, computes GCDs, normalizes results. This is unacceptable for real-time work.

The optimized implementation is 1.4 × slower than float on CPU and reaches parity on GPU. Here's how you get there.

Domain factorization

The single biggest win. Real workloads naturally cluster into domains where F is the same for all values:

Domain	Factor	Why
Transforms	F = 1	Positions are integer grid coordinates
Physics	F = 1000	Millisecond timestep, millimeter precision
Skinning	F = 32	Natural weight divisions, power of 2
UV/Texture	F = 256	8-bit texture precision, matches GPU formats
Particles	F = 1	Integer grid, discrete simulation

When F is the same for both operands, addition is just adding the numerators. No LCM computation. No normalization. When F is fixed for an entire domain, you don't even store it — it's implicit in the type. A "transform VFR" is just an i64. One integer. Eight bytes instead of twenty-four.

This alone drops your per-operation cost by 4-7 × .

Head-only evaluation

99.7% of operations resolve at depth zero. The recursive structure exists as capability but almost never fires. Your hot loop checks if R = 0 (which it almost always is) and takes the fast integer path. The rare operation that needs depth-1 precision pays for one pointer follow and one extra division. The even rarer depth-2 case pays a bit more. The average cost across all operations is nearly identical to pure integer arithmetic.

Bit-shift for power-of-2 factors

When F is a power of 2, division becomes a right shift. F = 32 means dividing by 32 is shifting right by 5. One CPU cycle instead of twenty to forty for integer division.

In profiled workloads, 69% of all factors are powers of 2. Skinning weights (F = 32), UV coordinates (F = 256), various internal scales — they're all shift-friendly. The hardware does the work in one cycle.

SIMD batching

When every value in a domain has the same F, you can process them in SIMD lanes with zero divergence. Eight particles at once, eight physics bodies at once, eight transform multiplications at once. No lane needs to check its factor or branch on normalization. They all do the same integer operation on different data.

On GPU, this means zero warp divergence. Every thread in a warp takes the same code path. The measured efficiency is 94% of theoretical SIMD throughput.

Reciprocal caching

A rigid body's mass doesn't change frame to frame. But you divide by it thousands of times per second — force/mass, impulse/mass, over and over. Compute the reciprocal once (swap V and F), store it, and multiply forever. Division disappears from your hot loop.

In physics simulations, this eliminates 98% of all divisions. Division is the most expensive integer operation (20-40 cycles). Replacing it with multiplication (3-4 cycles) across thousands of per-frame operations is a massive win.

The trajectory

These optimizations compound:

Implementation	Frame time	vs Float
Naive VFR (CPU)	42.3 ms	100 × slower
Generic optimization (CPU)	8.7 ms	2 × slower
Domain-specialized (CPU)	5.9 ms	1.4 × slower
GPU compute shaders	0.31 ms	faster for bulk work

The GPU number deserves emphasis. Modern GPUs have massive integer ALUs that the gaming world barely uses because everything runs through float pipelines. Domain-homogeneous VFR arithmetic — same factor, same operation, thousands of parallel threads — is the ideal GPU compute workload. 100,000 particles update in 0.02ms. 4,096 transforms compose in 0.05ms. All exact. All deterministic. All bit-identical across runs and platforms.

The rotation test from the opening: rotate a point by 120° three times with VFR. After three rotations, you're back at exactly [10, 0] with remainder [0, 0]. After three thousand rotations. After three million. The remainder returns to zero every three steps because the integers cancel exactly. No drift. Not now, not ever.

Appendix

A. Pell Convergent Table (√2)

Recurrence: V' = V + 2F, F' = V + F

V	F	$R = V^2 - 2F^2$	$(V/F)^2 - 2$
1	1	−1	−1/1
3	2	+1	+1/4
7	5	−1	−1/25
17	12	+1	+1/144
41	29	−1	−1/841
99	70	+1	+1/4900
239	169	−1	−1/28561
577	408	+1	+1/166464

R alternates ± 1 forever. Error shrinks as $1/F^2$. R never reaches 0.

B. The Five Domains

Domain	F	Storage	Division method	Use case
Transform	1	8 bytes (V only)	None needed	Positions, scales
Physics	1000	8 bytes (V only)	Integer /1000	Velocity, force, dt
Skinning	32	8 bytes (V only)	Right shift 5	Bone weights
UV/Texture	256	8 bytes (V only)	Right shift 8	Texture coordinates
Particles	1	8 bytes (V only)	None needed	Grid positions

Cross-domain conversion happens once per frame at well-defined boundaries. Overhead: under 0.3% of frame time.

C. Core Type Definitions (Zig)

zig

/// Full VFR — use when domain is mixed or unknown

const VFR = struct {

v: i64 = 0,

f: i64 = 1,

r: i64 = 0,

};

```
/// Transform domain — F=1 implicit, not stored
const TransformVFR = struct {
v: i64 = 0,
fn add(a: TransformVFR, b: TransformVFR) TransformVFR {
return .{ .v = a.v + b.v };
}
fn mul(a: TransformVFR, b: TransformVFR) TransformVFR {
return .{ .v = a.v * b.v };
}
};
/// Physics domain — F=1000 implicit
const PhysicsVFR = struct {
v: i64 = 0,
fn add(a: PhysicsVFR, b: PhysicsVFR) PhysicsVFR {
return .{ .v = a.v + b.v };
}
fn scale(self: PhysicsVFR, s: PhysicsVFR) PhysicsVFR {
return .{ .v = @divTrunc(self.v * s.v, 1000) };
}
};
/// Skinning domain — F=32 implicit, Lex-aligned
const SkinningVFR = struct {
v: i64 = 0,
fn applyWeight(weight: SkinningVFR, pos: TransformVFR) TransformVFR {
return .{ .v = (weight.v * pos.v) >> 5 };
}
};
/// 3D vector in transform domain
const Vec3Transform = struct {
x: TransformVFR = .{},
y: TransformVFR = .{},
```

z: TransformVFR = .{},

const ZERO = Vec3Transform{};

fn add(a: Vec3Transform, b: Vec3Transform) Vec3Transform {

```
return .{
    .x = .{ .v = a.x.v + b.x.v },
    .y = .{ .v = a.y.v + b.y.v },
    .z = .{ .v = a.z.v + b.z.v },
};
```

}

};

D. Key Identities

Pell relation: $V^2 - 2F^2 = R$, where $R = \pm 1$ for convergents of $\sqrt{2}$

Pell recurrence: V' = V + 2F, F' = V + F

VFR evaluation (depth-limited):

evaluate([V, F, R], depth):

if depth = 0 or R = 0: return V / F

return V/F + evaluate(R, depth - 1) / F

Addition (same factor): $[V_1, F, R_1] + [V_2, F, R_2] = [V_1 + V_2, F, R_1 + R_2]$

Multiplication: $[V_1, F_1, 0] \times [V_2, F_2, 0] = [V_1 \times V_2, F_1 \times F_2, 0]$

Domain conversion: To change factor from F_old to F_new, scale V by F_new / F_old

Verification: For any matrix M, check $M \times M^{-1} = I$ by binary integer equality on every entry. No epsilon.

[**CKS-MATH-128-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-128-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
MATH-0-2026

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026

[**CKS-MATH-127-2026**] The Taylor Continuum: $\mathbb{R}$ as an Unnamed Infinite Polynomial. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-127-2026

[**CKS-NEXT-1-2026**] Post-CKS. CKS is invalidated and falsified. Github: https://github.com/ghowland/cks/tree/main/papers/NEXT/CKS-NEXT-1-2026

[**CKS-LEX-12-2026**] Measurement Systems in the $\mathbb{Q}$-Substrate. Github: https://github.com/ghowland/cks/tree/main/paper LEX-12-2026